6 **(a)** State the conditions necessary for an object to have *simple harmonic motion.*

.....

.....

.....

..... [2]

(b) A long bar magnet hangs from one end of a spring as shown in Figure 6.1.

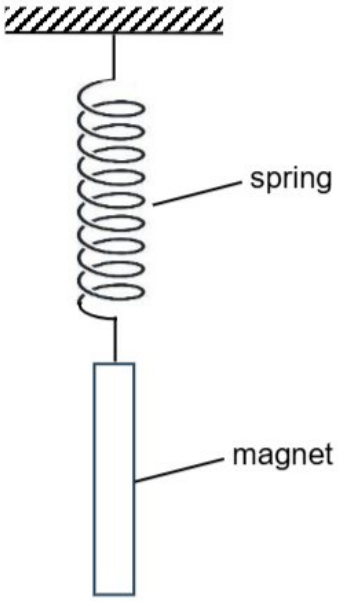


Figure 6.1

The magnet is displaced vertically downwards and then released. The subsequent vertical displacement x is found to vary with time as shown in Figure 6.2.

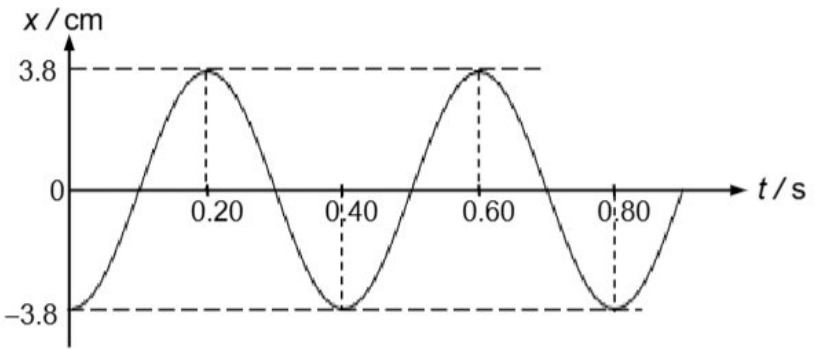


Figure 6.2

(i) Using values from Figure 6.2,

1. calculate the angular frequency ω of the oscillation of the magnet.

ω = ... rad s^{-1} [2]

2. determine the acceleration of the magnet when the displacement is zero and when the displacement is at its maximum.

acceleration when displacement is zero = ... m s^{-2}

acceleration when displacement is maximum = ... m s^{-2}

[3]

(ii) On Figure 6.3, sketch the graph of the variation with displacement of the velocity of the magnet. Label the axes with appropriate values.

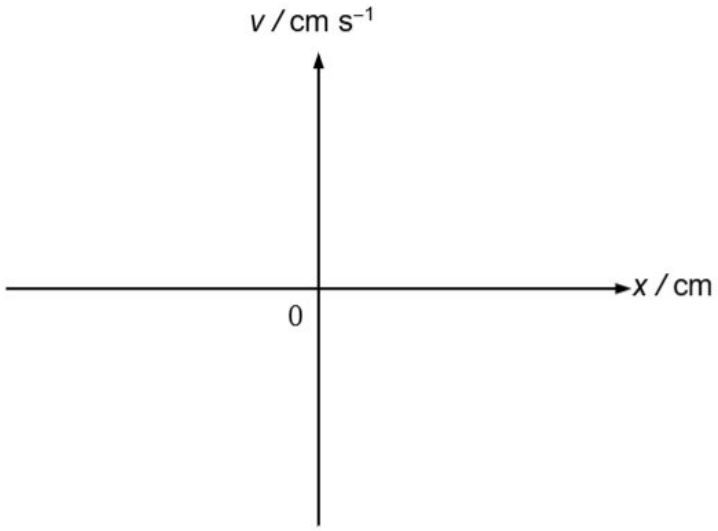


Figure 6.3

[2]

(iii) Determine the distance of the magnet from the equilibrium position when its speed is half of the maximum speed.

distance = ... cm [3]

(c) One pole of the magnet is now placed inside a coil of wire as shown in Figure 6.4. The magnet is given a small displacement and is then released so that the north pole of the magnet oscillates within the coil.

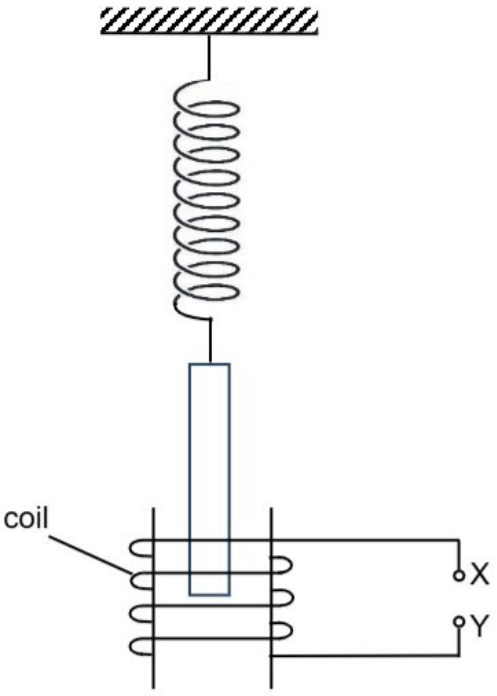


Figure 6.4

An induced e.m.f. is measured across the terminals XY of the coil.

(i) On Figure 6.5, sketch the variation with time of the induced e.m.f. as the magnet oscillates according to the displacement-time graph of Figure 6.2.

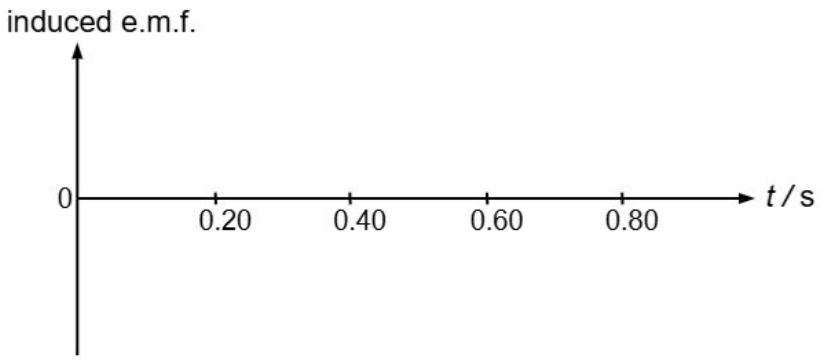


Figure 6.5

[2]

(ii) Using the principles of electromagnetic induction and with reference to Figure 6.2, explain the variation of the induced e.m.f. drawn in **6(c)(i)**.

.....

.....

.....

.....

.....

..... [3]

(iii) State and explain the change, if any, that occurs in the amplitude of the oscillations of the magnet when a resistor is connected across XY.

.....

.....

.....

.....

.....

.....

..... [3]

[Total: 20]